Exercise 13.2

1. A researcher collected data on number of deaths from Horse-Ricks in Russian Army crops over to years. The table is as follows:

No. of death	**0**	**1**	**2**	**3**	**4**	**5**	**6**
Frequency	**60**	**50**	**87**	**40**	**32**	**15**	**10**

Find the relative frequency of the given data.

No. of Deaths	Frequency	Relative Frequency
0	60	$\frac{60}{294}=\frac{30}{147}$
1	50	$\frac{50}{294}=\frac{25}{147}$
2	87	$\frac{87}{294}=\frac{29}{98}$
3	40	$\frac{40}{294}=\frac{20}{147}$
4	32	$\frac{32}{294}=\frac{16}{147}$
5	15	$\frac{15}{294}=\frac{5}{98}$
6	10	$\frac{10}{294}=\frac{5}{147}$
Total	$\sum f=294$	

2. The frequency of defective products in 750 samples are shown in the following table.

No. of defectives per sample	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
No. of sample	**120**	**140**	**94**	**85**	**105**	**50**	**40**	**66**	**50**

Find the relative frequency for the given table.

No. of Defectives	No. of Samples	Relative Frequency
0	120	$\frac{120}{750}=\frac{4}{25}$
1	140	$\frac{140}{750}=\frac{14}{75}$
2	94	$\frac{94}{750}=\frac{47}{375}$
3	85	$\frac{85}{750}=\frac{17}{150}$
4	105	$\frac{105}{750}=\frac{7}{50}$
5	50	$\frac{50}{750}=\frac{1}{15}$
6	40	$\frac{40}{750}=\frac{4}{75}$
7	66	$\frac{66}{750}=\frac{11}{125}$
8	50	$\frac{50}{750}=\frac{1}{15}$
Total	$\sum f=750$	

3. A quiz competition on general knowledge is conducted. The number of corrected answers out of 5 questions for 100 sets of questions is given below.

X	0	1	2	3	4	5
f	10	23	15	25	18	9

Find the relative frequency.

X	f	Relative Frequency
0	10	$\frac{10}{100}=\frac{1}{10}$
1	23	$\frac{23}{100}=\frac{23}{100}$
2	15	$\frac{15}{100}=\frac{3}{20}$
3	25	$\frac{25}{100}=\frac{1}{4}$
4	18	$\frac{18}{100}=\frac{9}{50}$
5	9	$\frac{9}{100}=\frac{9}{100}$
Total	$\sum f=100$	

4. A survey was conducted from the 108 students of a class and asked about their favourite food. The responses are as under:

Name of food item	Biryani	Fresh Juice	Chicken	Bar. B.Q	Sweets
No. of students	40	07	21	15	25

$$Total\ students=40+7+21+15+25=108$$

(i) how many percentages of students like biryani?

$$Percentage=\frac{No.\ of\ students\ for\ Biryani}{Total\ students}\times 100$$

$$Percentage=\frac{40}{108}\times 100$$

$$\boldsymbol{Percentage=37.04\%}$$

(ii) how many percentages of students like chicken?

$$Percentage=\frac{No.\ of\ students\ for\ Chicken}{Total\ students}\times 100$$

$$Percentage=\frac{21}{108}\times 100$$

$$\boldsymbol{Percentage=19.44\%}$$

(iii) which food is the least like by the students?

From the table, $\boldsymbol{Fresh\ Juice\ =\ 7\ students}$ is the lowest.

(iv) which food is the most prefer by the students?

From the table, $\boldsymbol{Biryani\ =\ 40\ students}$ is the highest.

5. In 500 trials of a thrown of two dice, what is expected frequency that the sum will be greater than 8?

When two dice are thrown, the sample space will be:

$$\begin{aligned}S=\{&(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),\\&(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),\\&(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),\\&(4,1),(4,2),(4,3),(4,4),(4,5),(4,6),\\&(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),\\&(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}\end{aligned}$$

$$n(S)=36$$

Let A be the event getting the sum greater than 8.

$$A = \{(3,6), (4,5), (5,4), (6,3), (4,6), (5,5), (6,4), (5,6), (6,5), (6,6)\}$$

$$n(A) = 10$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{10}{36}$$

$$P(A) = \frac{5}{18}$$

Now by using formula

$$Expected\ frequency = Total\ number\ of\ trials \times Probability\ of\ the\ event$$

$$E = N \times P(A)$$

$$E = 500 \times \frac{5}{18}$$

$$\boldsymbol{E = 138.9}$$

6. What is the expectation of a person who is to get Rs. 120 if he obtains at least 2 heads in single toss of three coins?

When 3 coins are tossed, the sample space will be:

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$$n(S) = 8$$

Let A the event of getting at least 2 heads in single toss

$$A = \{HHT, HTH, THH, HHH\}$$

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{4}{8}$$

$$P(A) = \frac{1}{2}$$

Now by using formula

$$Expected\ frequency = Total\ number\ of\ trials \times Probability\ of\ the\ event$$

$$E = N \times P(A)$$

$$E = 120 \times \frac{1}{2}$$

$$\boldsymbol{E = 60}$$

7. Find the expected frequencies of the given data:

x	0	1	2	3	4	5	6
$P(x)$	0.11	0.21	0.17	0.18	0.09	0.17	0.07

if the experiment is repeated 200 times.

x	$P(x)$	$E = N \times P(x)$ $E = 200 \times P(x)$
0	0.11	$200 \times 0.11 = 22$
1	0.21	$200 \times 0.21 = 42$
2	0.17	$200 \times 0.17 = 34$
3	0.18	$200 \times 0.18 = 36$
4	0.09	$200 \times 0.09 = 18$
5	0.17	$200 \times 0.17 = 34$
6	0.07	$200 \times 0.07 = 14$
Total	**1.00**	**200**

8. The probability of getting 5 sixes while tossing six dice is $\frac{2}{5}$, the dice is rolled 200 times. How many times would you expect it to show 5 sixes?

$$Probability\ of\ getting\ 5\ sixes = P(5\ sixes) = \frac{2}{5}$$

Now by using formula

$$Expected\ frequency = Total\ number\ of\ trials \times Probability\ of\ the\ event$$

$$E = N \times P(5\ sixes)$$

$$E = 200 \times \frac{2}{5}$$

$$\boldsymbol{E = 80\ times}$$